

Assignment 2: (Read and Run Code Assignment)

9/13/26

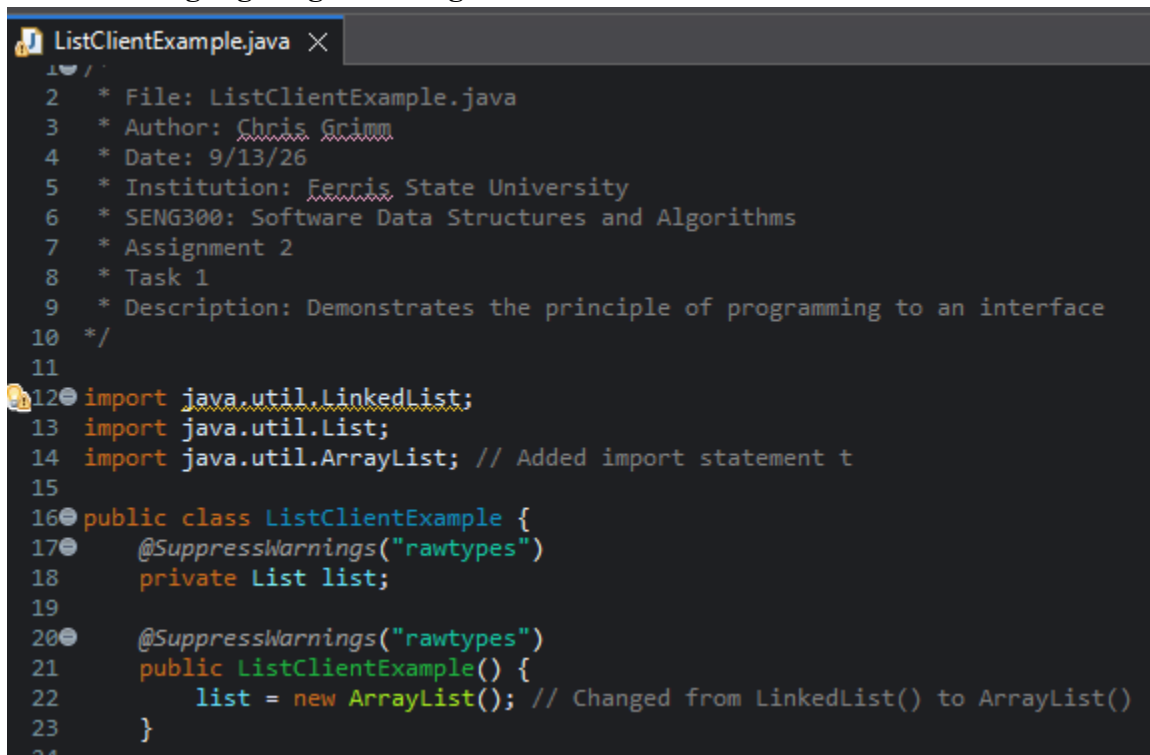
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Details

Task 1) Testing concepts from the lecture

Run the ListClientExample.java file from the thinkData project.

Change the data structure used from LinkedList to ArrayList, Note that this is one change in the whole code. (in addition to adding 1 import statement). A) submit the code file + screenshot highlighting the changed statements.



```
1  //
2  * File: ListClientExample.java
3  * Author: Chris Grimm
4  * Date: 9/13/26
5  * Institution: Ferris State University
6  * SENG300: Software Data Structures and Algorithms
7  * Assignment 2
8  * Task 1
9  * Description: Demonstrates the principle of programming to an interface
10 */
11
12 import java.util.LinkedList;
13 import java.util.List;
14 import java.util.ArrayList; // Added import statement t
15
16 public class ListClientExample {
17     @SuppressWarnings("rawtypes")
18     private List list;
19
20     @SuppressWarnings("rawtypes")
21     public ListClientExample() {
22         list = new ArrayList(); // Changed from LinkedList() to ArrayList()
23     }
24 }
```

B) Why is that change in one line instead of many throughout the code?

The variable list is declared as the interface type List.

C) if you are building your code in python, you can apply the code in the linkedlist articles below instead.

Task 2)

Review and run the examples of the following 2 articles

1) using LinkedLists in Your java code

<https://beginnersbook.com/2013/12/linkedList-in-java-with-example/>

2) Implementing a linked list of your own

<https://www.geeksforgeeks.org/implementing-a-linked-list-in-java-using-class/>

Alternatively, you can use the code in the Java zybook section 4.6

for students using Python:

1) the zybook has detailed explanation in section 7.1 and 7.2 in addition to examples

2) As a really nice additional resource: you can review this

link <https://realpython.com/linked-lists-python/>

with a focus on running these articles:

- [Understanding Linked Lists](#)
 - [Main Concepts](#)
 - [Practical Applications](#)
 - [Performance Comparison: Lists vs Linked Lists](#)
- [Introducing collections.deque](#)
 - [How to Use collections.deque](#)
- [Implementing Your Own Linked List](#)
 - [How to Create a Linked List](#)
 - [How to Traverse a Linked List](#)
 - [How to Insert a New Node](#)
 - [How to Remove a Node](#)

Your submission should contain any questions you have about LinkedLists after running these examples.

Task 3) if you were running a DMV for a small village. You had to implement a program to store car information where an object stores the information for 1 car, which data structure would you use in the following scenarios:

A) number of cars is more or less stable at around a 1000 cars

ArrayList as the number of cars will not grow.

B) number of cars keeps changing so you cannot estimate an average (people move out and in the city quite a lot).

LinkedList as the numbers of cars can change.